Jinwon Lee

Agent Platforms · Applied AI · Reliability

Website

https://jinwonlee.github.io/

This portfolio summarizes the agent platform and reliability engineering work that takes enterprise and user-facing applications from requirements through implementation, rollout, and production stabilization. Each project is organized around the problem context, my role, engineering and delivery focus, outcomes, and capability signals.

Agent Platforms

Agent runtime integration, self-service application delivery, and existing LLM Gateway API connectivity

Governed Delivery

Authentication, authorization, security controls, automatic domains, and enterprise governance

Reliability

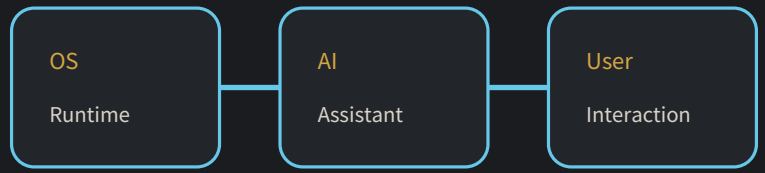
Observability, incident response, rollout readiness, and production stabilization

Cloud Runtime

AWS EKS, Azure, OpenShift, Kubernetes, Helm, ArgoCD, and GitOps

Project Details

Problem, role, engineering/delivery focus, outcome, and capability point



01. Siri-like AI Assistant on TmaxOS

AI assistant with system-level integration and natural user experience.

Problem	The OS environment needed an AI assistant that could connect user interaction with system-level capabilities.
Role	Built AI inference and preprocessing flows and integrated them with TmaxOS-level behavior.
Engineering	Async architecture, system integration, AI model inference, and data preprocessing.
Outcome	Delivered an AI assistant experience with smoother interaction through asynchronous design.
Capability	Experience integrating AI functionality with OS-level behavior, asynchronous processing, and user interaction.

Python · AI/ML · TmaxOS · Async Programming

02. High-throughput Packet Mirroring Application

Real-time packet mirroring application that forwarded up to 130,000 packets/sec under constrained resources.

Problem	Network/application teams needed real-time packet forwarding under limited system resources.
Role	Implemented the packet mirroring application and tuned it for resource-constrained operation.
Engineering	High-throughput I/O, Linux runtime behavior, resource limits, and real-time data forwarding.
Outcome	Delivered a system component capable of forwarding up to 130,000 packets/sec.
Capability	Experience optimizing Linux runtime behavior and high-throughput data movement under resource constraints.

Linux · Networking · System Software · Performance

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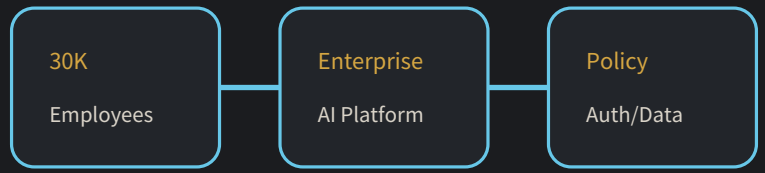

03. Company-wide AI Chatbot Platform (30K Users)

Enterprise AI chatbot platform serving 30,000 employees with Kubernetes-based application delivery, reliability, and data protection.

Problem	Provide a reliable internal AI assistant platform for tens of thousands of employees.
Role	Led platform and application reliability across development, staging, and production, connecting releases with security and operational requirements.
Engineering	Kubernetes operations, Helm/Helmfile IaC, AI service delivery, authentication, and data protection compliance.
Outcome	Supported a company-wide AI service with production-grade operating standards.
Capability	Experience leading application delivery and reliability for tens of thousands of internal users.

Kubernetes · Internal Cluster · Application Delivery

04. B2C Generative AI Service (Korea & UK)

Microsoft Azure-based B2C generative AI service launched in Korea and the UK with initial production operations and incident response.

Problem	Launch a consumer-facing generative AI service in Korea and the UK and stabilize its initial production operations.
Role	Delivered platform/SRE responsibilities for Korea and UK launch readiness, initial production operations, incident response, and early stabilization.
Engineering	Azure and Kubernetes operations, rollout readiness, production stabilization, and incident response.
Outcome	Delivered Korea and UK launch readiness and early production stabilization, completing the engagement in June 2026.
Capability	Experience delivering Korea and UK launch readiness, production stabilization, and incident response for a consumer generative AI service.

Azure · Kubernetes · Generative AI · SRE

Project Details

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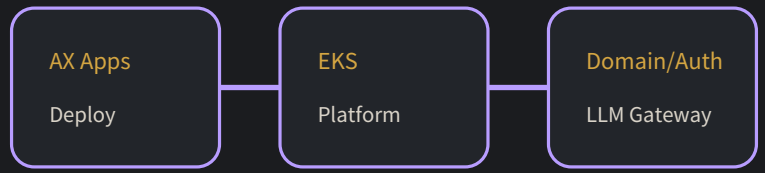


05. Internal AX Agent Platform

AWS EKS-based platform for deploying and operating AX applications with automated domains, access control, governance, and LLM Gateway integration.

Problem	Enable teams to deploy AX applications quickly while enforcing enterprise access control, security, and AI governance.
Role	Owned end-to-end platform delivery from architecture and implementation through deployment and operation, including authentication, authorization, security, and internal governance.
Engineering	AWS EKS, AX Agent runtime adaptation, automated domain provisioning, access control, policy enforcement, and client integration with the existing LLM Gateway.
Outcome	Created a self-service path for governed AX application deployment with automatic domains and authorized-only access.
Capability	Experience building a governed self-service agent platform with runtime compatibility, access control, and enterprise LLM integration.

AWS · EKS · Kubernetes · AuthN/AuthZ · Governance

06. Centralized Monitoring & Observability Stack

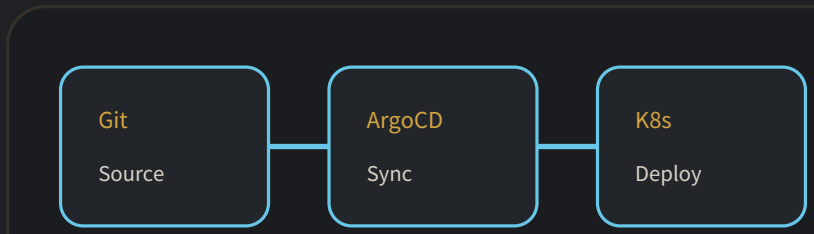
Centralized observability stack for system metrics, container runtime metrics, dashboards, and alerts.

Problem	Production services needed shared visibility into system behavior and container runtime health.
Role	Built the monitoring path and dashboards used for platform and service operations.
Engineering	Prometheus metrics, Grafana dashboards, OpenTelemetry signals, Loki logging, and alert rules.
Outcome	Improved operational visibility for incident detection and runtime analysis.
Capability	Experience using metrics, logs, and traces for incident detection and runtime analysis.

Grafana · Prometheus · OpenTelemetry · Loki

Project Details

Problem, role, engineering/delivery focus, outcome, and capability point



07. GitOps CI/CD Pipeline with ArgoCD

GitOps-based delivery pipelines for containerized AI/platform services with Python automation and DevSecOps checks.

Problem	Containerized AI/platform services needed repeatable, auditable, and secure deployment workflows.
Role	Designed delivery automation and connected AI service deployment pipelines with security verification steps.
Engineering	ArgoCD GitOps, GitHub Actions, Docker image handling, Kubernetes deployment, and Python glue automation.
Outcome	Improved deployment consistency and reduced reliance on manual release operations.
Capability	Experience connecting deployment consistency, security checks, and operations automation for containerized services.

ArgoCD · GitHub Actions · Docker · Kubernetes

08. Kubernetes Cluster Automation

One-click Kubernetes cluster provisioning for repeatable developer and test environments.

Problem	Cloud developers needed repeatable Kubernetes environments without manual setup overhead.
Role	Designed and implemented provisioning automation for Kubernetes clusters and shared platform modules.
Engineering	Bash-based provisioning and automation, reusable cluster modules, and developer environment standardization.
Outcome	Reduced manual operations and improved environment consistency for cloud development workflows.
Capability	Experience standardizing infrastructure and automating repeatable development and test environments.

Kubernetes · Bash

Project Details

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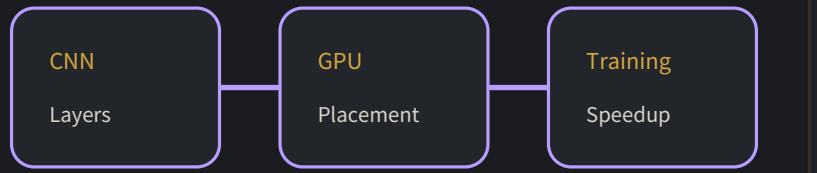

09. Distributed CNN Training System

Research on accelerated CNN distributed deep learning with automatic resource-aware layer placement.

Problem	Distributed CNN training suffered from network bottlenecks and inefficient layer placement.
Role	Contributed to research on resource-aware placement for distributed TensorFlow/CNN training.
Engineering	Distributed systems, model parallelism, resource-aware scheduling, TensorFlow, and GPU-aware workloads.
Outcome	Published research artifacts including USENIX NSDI 2019 poster and arXiv paper; reported up to 2.3x training speedup.
Capability	Research experience on network bottlenecks, resource placement, and performance improvement for distributed training workloads.

TensorFlow · Distributed Systems · AI Workloads · Python

10. LLM-Assisted Cluster Monitoring

Personal project for internal cluster operations: error detection, LLM analysis, and email reports with suggested remediation steps.

Problem	Cluster errors need interpretation and practical troubleshooting guidance, beyond a basic alert.
Role	Built the monitoring and reporting workflow as a personal project for internal cluster operations.
Engineering	Connected error detection, LLM-based issue analysis, and email delivery of problem summaries and remediation suggestions.
Outcome	Delivered email reports containing the detected problem, LLM analysis, and suggested steps for the operator to review.
Capability	Applies LLM analysis to cluster operations and operator-facing incident reporting.

LLM · Cluster Monitoring · Email Alerts

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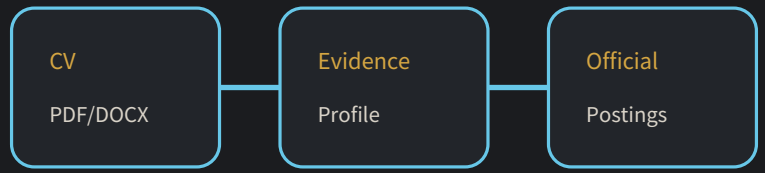

11. OfferNow

Personal project: a CV-based job recommendation service for Korean IT roles that shows the CV evidence behind every recommended official posting.

Problem	Job seekers have to read many postings to see which ones actually fit their experience, and keyword matching gives no reason they can check.
Role	Personal project: product definition, backend, mobile and web client, data collection, and deployment.
Engineering	Expo/React Native client for web, iOS, and Android; FastAPI backend; PostgreSQL with row-level security and anonymous 24-hour sessions; a daily collector with a durable task queue; rule-based matching; provider-neutral LLM calls with usage limits and cost logging; GitHub Actions CI.
Outcome	Deployed at heyoffernow.ai behind an operator-only gate ahead of a beta; collects official postings from 50+ companies.
Capability	Builds a full AI product end to end, from data collection and privacy controls to LLM integration, cost limits, and deployment.

Expo FastAPI PostgreSQL LLM Docker

Links and References

Detailed projects and the latest CV are available on the online portfolio.

Website

https://jinwonlee.github.io/

GitHub

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Awards & Activities

- Finalist, SK hynix AI Hackathon 2026 (Nov. 2026): Advanced through two online qualifying rounds to the offline finals (individual entry).

Publications

- Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement — USENIX NSDI 2019 (2019)
- Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement — arXiv 2019 (2019)
- Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism — ACM EuroSys 2018 (2018)

Additional experience: contributed to LLM evaluation work for machine translation.

This document is a portfolio summary based on publicly shareable career and project information. It does not include confidential implementation details or non-public information.